Example 4.2

In the Fig.4.35 $ABCD$ is a parallelogram, P and Q are the mid-points of sides AB and DC respectively. Show that $APCQ$ is a parallelogram.

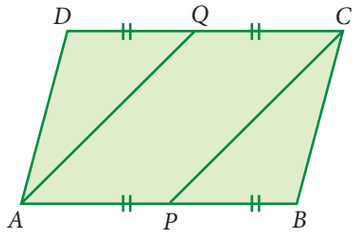


Fig. 4.35

Solution

Since P and Q are the mid points of

AB and DC respectively

Therefore $AP = \frac{1}{2} AB$ and

$$QC = \frac{1}{2} DC \qquad (1)$$

But $AB = DC$ (Opposite sides of a parallelogram are equal)

$$\Longrightarrow \quad \frac{1}{2} AB = \frac{1}{2} DC$$

$$\Longrightarrow \quad AP = QC \qquad (2)$$

Also , $AB \parallel DC$

$$\Longrightarrow \quad AP \parallel QC \qquad (3)\ [\because\ ABCD \text{ is a parallelogram}]$$

Thus , in quadrilateral $APCQ$ we have $AP = QC$ and $AP \parallel QC$ [from (2) and (3)]

Hence , quadrilateral $APCQ$ is a parallelogram.

Example 4.3

$ABCD$ is a parallelogram Fig.4.36 such that $\angle BAD = 120^{\circ}$ and AC bisects $\angle BAD$ show that $ABCD$ is a rhombus.

Solution

Given $\angle BAD = 120°$ and AC bisects $\angle BAD$

$$\angle BAC = \frac{1}{2} \times 120° = 60°$$

$$\angle 1 = \angle 2 = 60°$$

$AD \parallel BC$ and AC is the traversal

$$\angle 2 = \angle 4 = 60°$$

ΔABC is isosceles triangle $[\because \angle 1 = \angle 4 = 60\ °]$

$$\Longrightarrow \quad AB = BC$$

Parallelogram $ABCD$ is a rhombus.

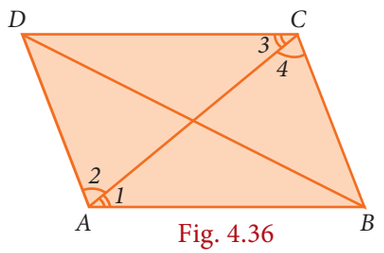


Fig. 4.36

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